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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COMPILER DESIGN (24CS2235F)

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TITLE OF THE PROJECT		Lexical Analyzer Using LEX Tool	
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ABSTRACT

The Lexical Analyzer Using LEX Tool is a Compiler Design project that focuses on the implementation of the lexical analysis phase, which is responsible for processing source code and identifying its basic components. Lexical analysis is an important initial phase of a compiler because it converts the input program into a sequence of meaningful tokens that can be further processed by the subsequent phases of compilation. This project demonstrates how lexical analysis can be performed using LEX/FLEX with C and how regular expressions can be used to define and recognize different types of tokens.

The proposed system accepts source code as input and scans it character by character to identify different lexical elements such as keywords, identifiers, operators, and other symbols. It also removes unnecessary comments and whitespace from the input, helping to simplify the source code before further processing. The analyzer identifies and categorizes the recognized tokens based on predefined patterns and regular expressions. In addition to token identification, the system maintains a count of the tokens encountered during the analysis, providing useful information about the lexical structure of the given program.

Another important feature of the project is the generation of a symbol table, which stores information related to the identified tokens and provides an organized representation of the lexical elements found in the source code. Through these features, the project demonstrates the practical working of a lexical analyzer and its role in a compiler. The implementation provides a simple way to understand concepts such as tokens, regular expressions, finite automata, and lexical analysis.

The project is designed to provide a clear and practical understanding of the first stage of compiler construction while keeping the implementation simple and manageable. By using LEX/FLEX and C, the system demonstrates how predefined lexical rules can automatically recognize and classify elements of a source program. Overall, the project helps in understanding how raw source code is transformed into structured tokens that can be passed to later stages of the compiler, making it a suitable and effective implementation of lexical analysis concepts.